Automating Segment Routing Traffic Engineering

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Today

- Abstract: In this talk, we will present a brief introduction to Segment Routing, followed by some of the recent technology developments in this area. We will cover Dynamic Paths, Policies, Flexible Algorithms for dynamic path computation, and how we can bring these technologies together in an automated transport SDN solution.

1. Segment Routing Refresher
2. Current Developments in Segment Routing Traffic Engineering
3. How we can Automate and Program Infrastructure using a Controller

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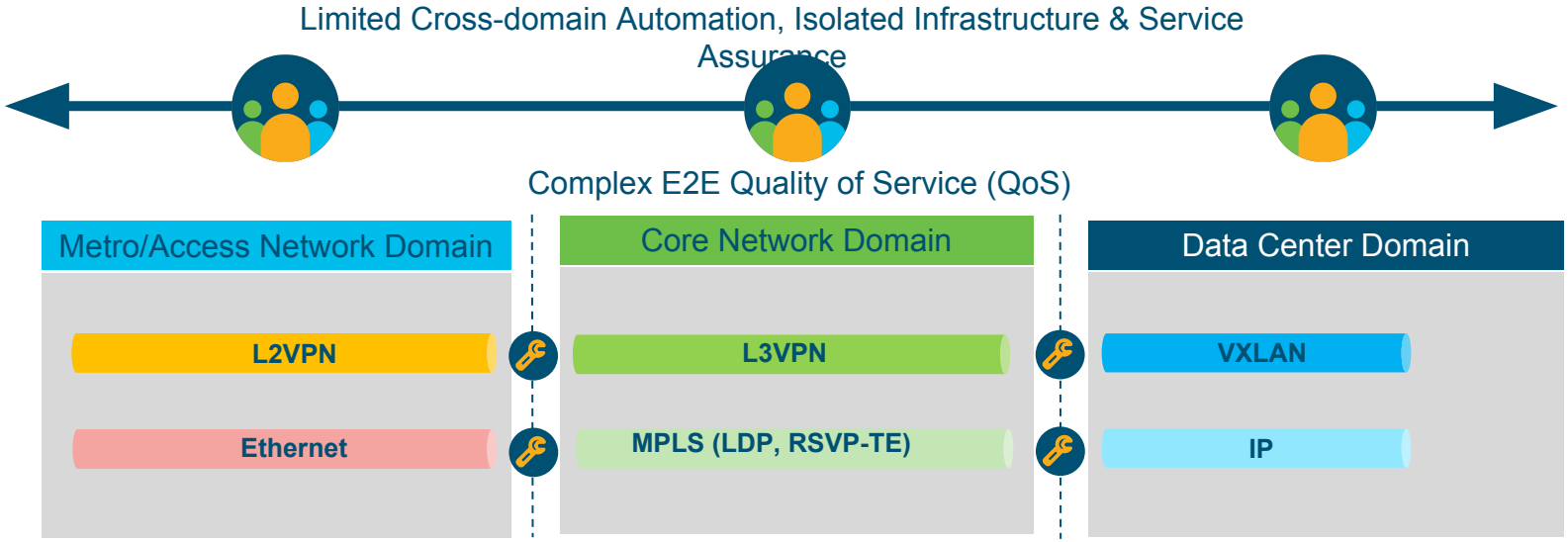
Today: Operational Domains and Layers of Protocols



Automated end-to-end service provisioning can be tricky

- Multiple network domains under different management teams
- Manual operations
- Different types of underlay and overlay networks

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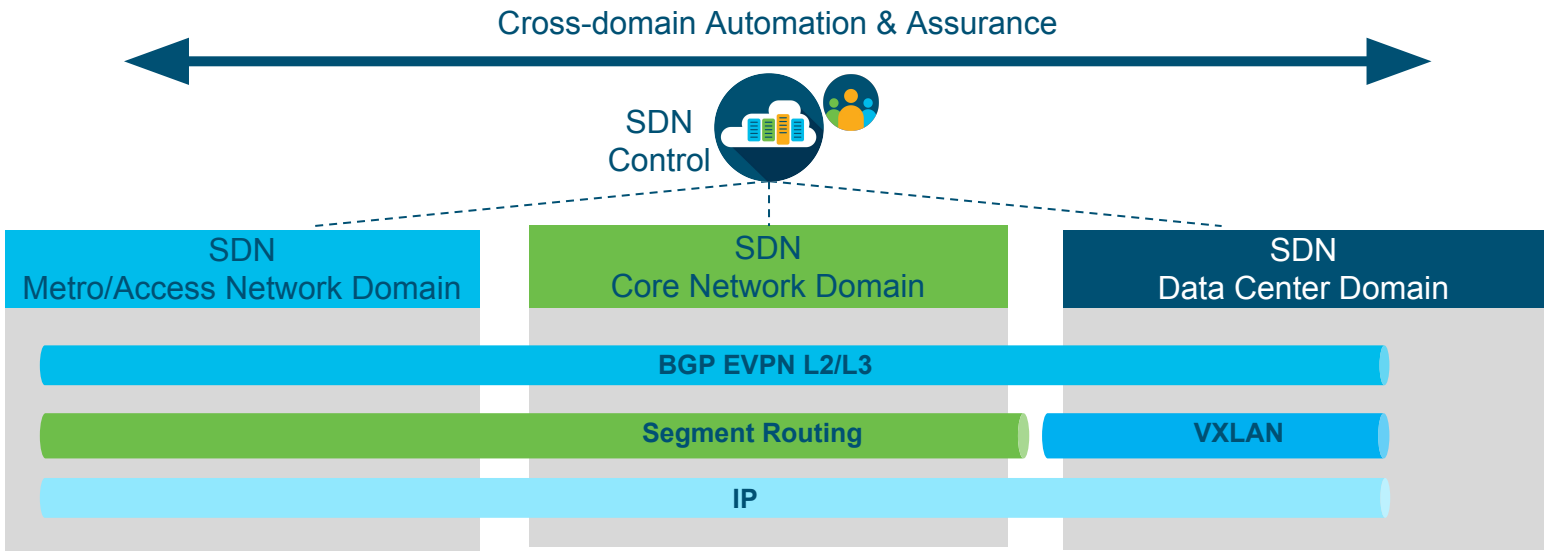
Target: End-to-End and Domain automation



End-to-end service provisioning is more easily automated

- Common underlay and overlay network technologies
- Automated underlay and overlay operations
- Seamless integration between domains

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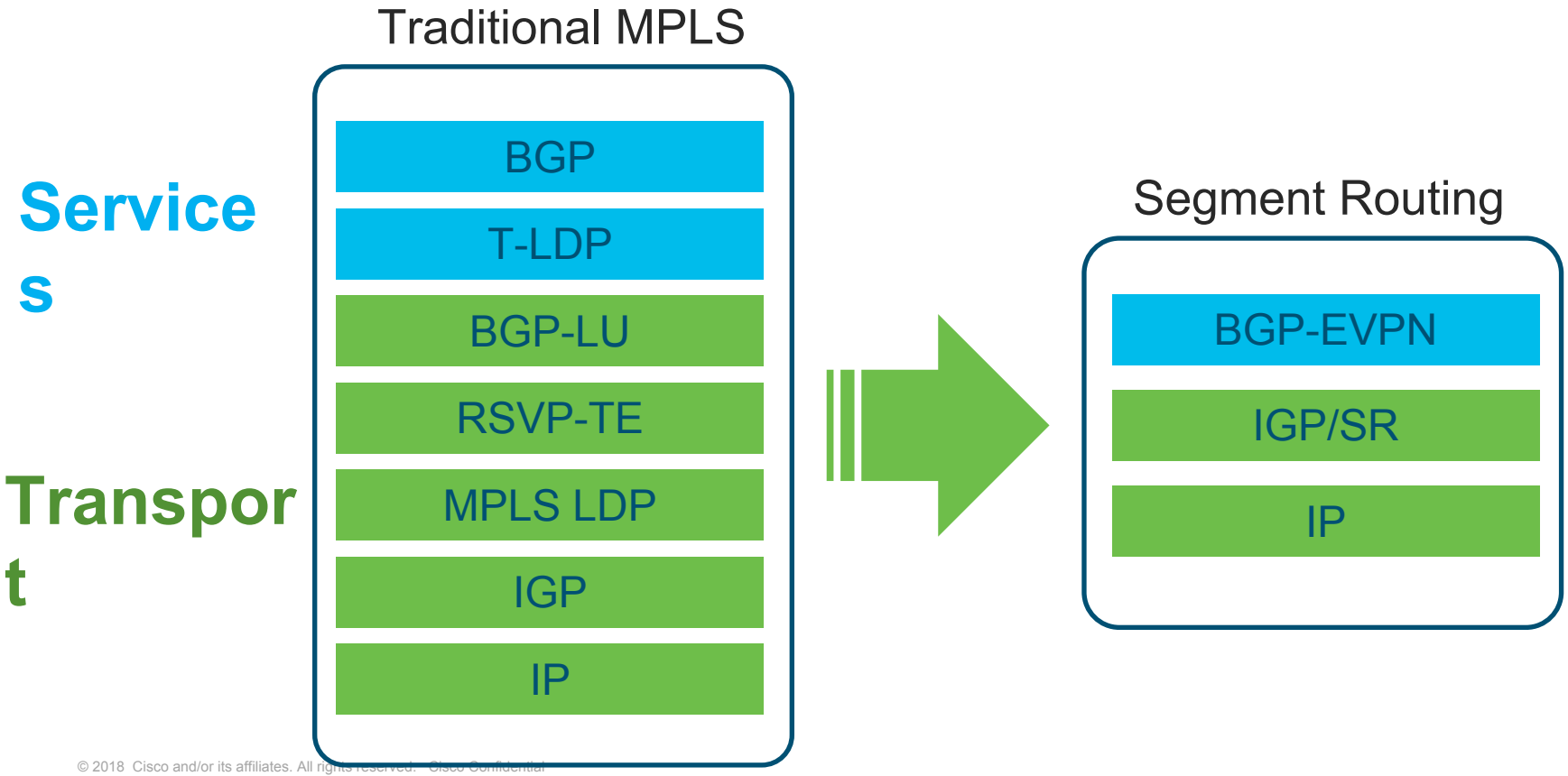
Segment Routing Protocol Efficiency



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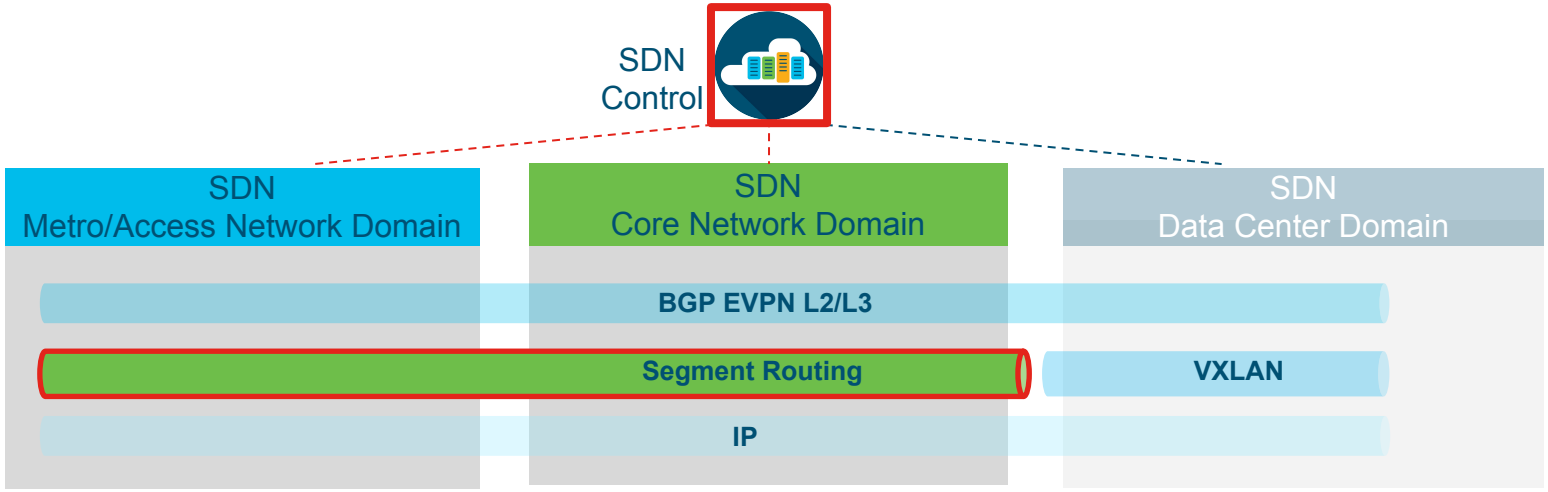
Focus for Today



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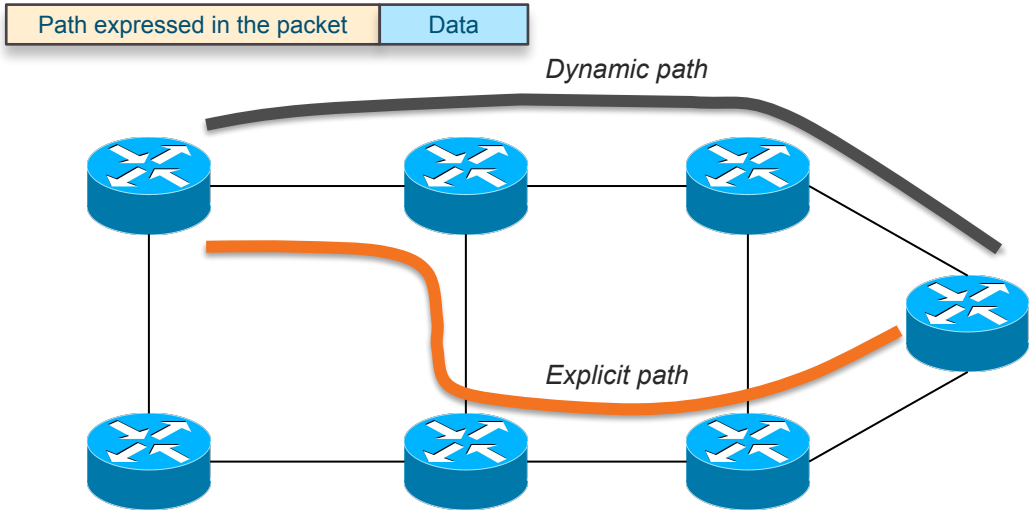
Segment Routing Refresher

Segment Routing – Source Routing



Data Plane

- MPLS (segment labels)
- IPv6 (+ SR extension header)

Control Plane

- Routing protocols with extensions (IS-IS,OSPF, BGP)
- SDN controller (BGP , PCEP, NETCONF/YANG)

Path options

- Dynamic (SPF)
- Explicit (expressed in the packet)

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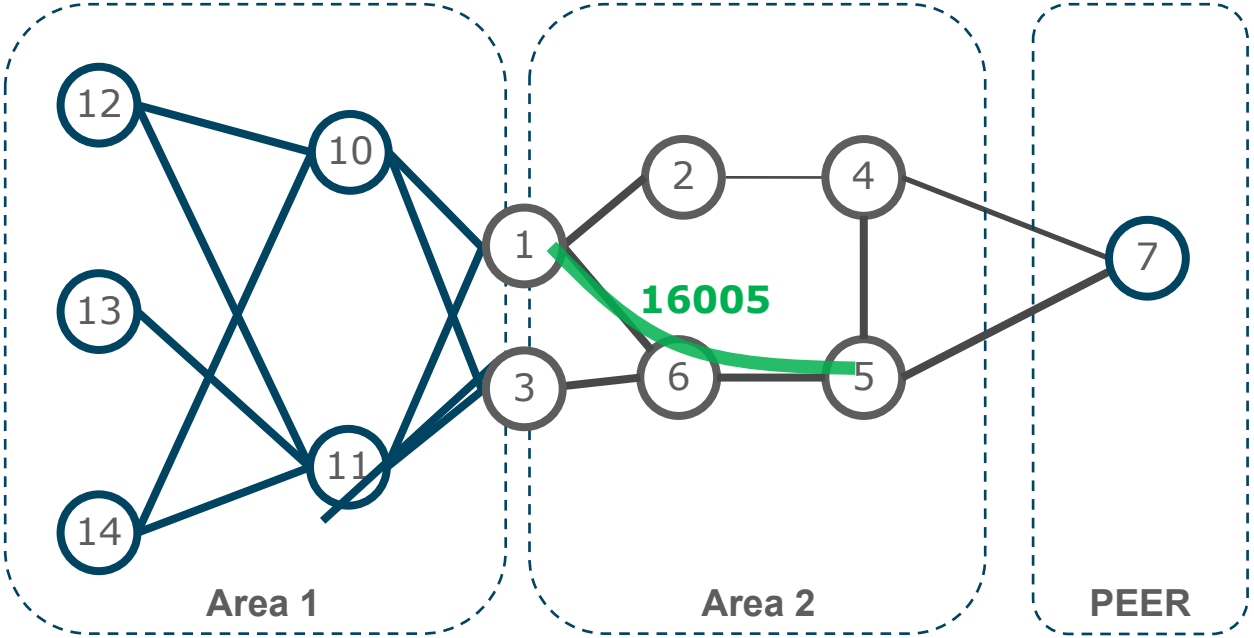
What is a Segment? - Prefix Segment

- Shortest path to the IGP prefix
- SID 16000 + Index
- Distributed in ISIS/OSPF



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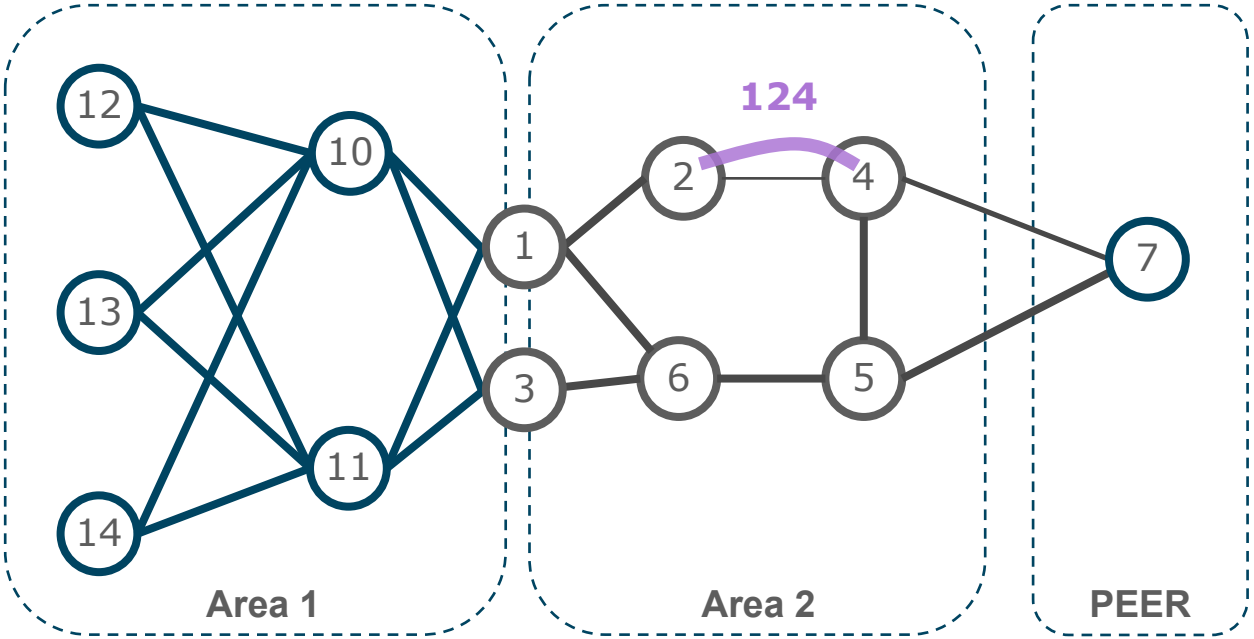
What is a Segment? - Adjacency Segment

- Forward on the IGP adjacency
- SID 1XY
  - From X
  - To Y
- Distributed in ISIS/OSPF



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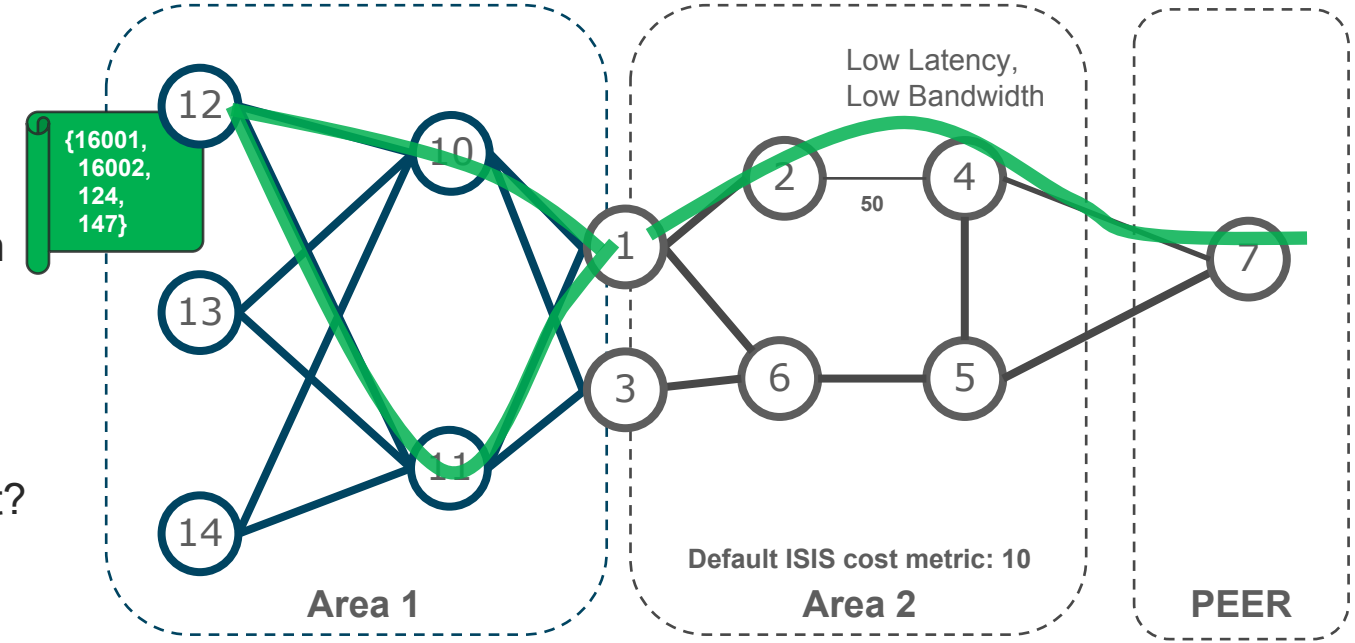
Creating the Segment List

- We can combine SIDs to form a *Segment List.*
  - 16001
  - 16002
  - 124
  - 147
- The *list* is encoded in the IP packet header
- How to create a dynamic segment list?



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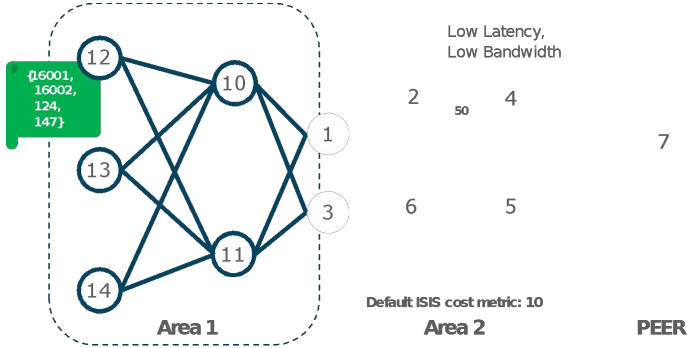
Distributed vs Centralised Path Computation

- There are two possibilities to compute the dynamic path for an SR Policy:
  - Head-end computes the path itself (“distributed”)
  - Head-end requests SR PCE to compute the path (“centralised”)
- By default, dynamic paths are computed by the head-end
- Head-end uses an external PCE when local computation is not possible
  - An external PCE is required if more information is needed than is available on a head-end; e.g. multi-area/domain paths, or disjoint paths from different head-ends



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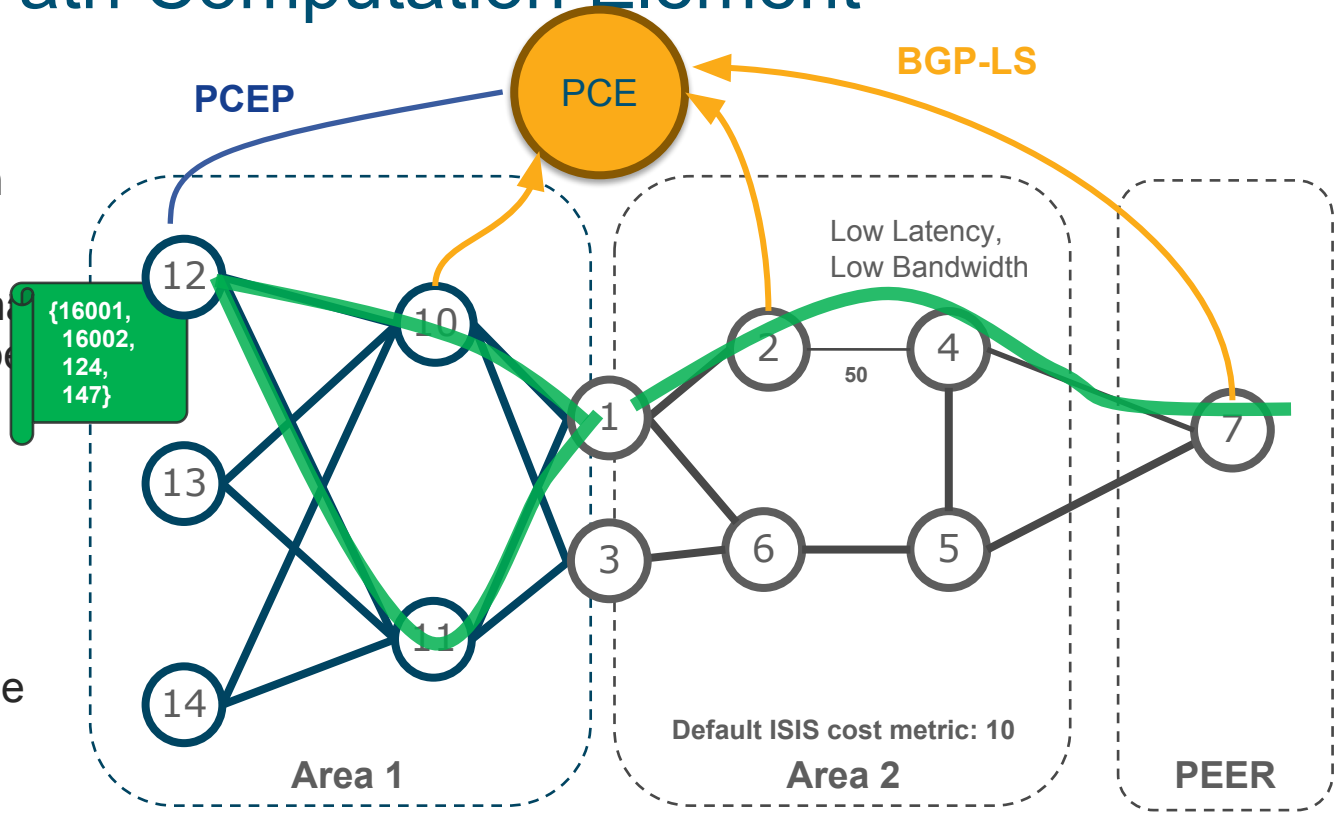
Centralised Path Computation Element

- The head-end is configured to use the PCE for dynamic path computation
- The PCE computes that the green path can be encoded as
  - 16001
  - 16002
  - 124
  - 147
- The PCE programs the path to the head-end



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Segment Routing in the IETF

- IETF standardization in SPRING working group
- First RFC document - RFC 7855 (May 2016)
- Protocol extensions progressing in multiple groups
  - **IS-IS**
  - **OSPF**
  - PCE
  - IDR
  - 6MAN
  - BESS
- Broad vendor support
- **MPLS** and IPv6 Data Planes

Sample IETF Documents
Problem Statement and Requirements (RFC 7855)
Segment Routing Architecture (draft-ietf-spring-segment-routing)
IPv6 SPRING Use Cases (draft-ietf-spring-ipv6-use-cases)
Segment Routing with MPLS data plane (draft-ietf-spring-segment-routing-mpls)
Topology Independent Fast Reroute using Segment Routing (draft-bashandy-rtgwg-segment-routing-ti-lfa)
IS-IS Extensions for Segment Routing (draft-ietf-isis-segment-routing-extensions)
OSPF Extensions for Segment Routing (draft-ietf-ospf-segment-routing-extensions)
PCEP Extensions for Segment Routing (draft-ietf-pce-segment-routing)

And many more drafts in progress

Current Developments in Segment Routing

A New Model for Traffic Engineering

- Simple, Automated and Scalable
  - No core state: state in the packet header
  - No tunnel interface: “SR Policy”
  - No head-end path configuration: on-demand policy instantiation
  - No head-end policy-based routing: automated steering

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Automatic SR TE Path Creation: Using SR Policy & On-Demand Next Hop (ODN)

What it Does

Automatic SR TE Path Creation

Edge router automatically computes a path to the remote service endpoint

The path can either be for simple best effort reachability or for reachability with SLA contract

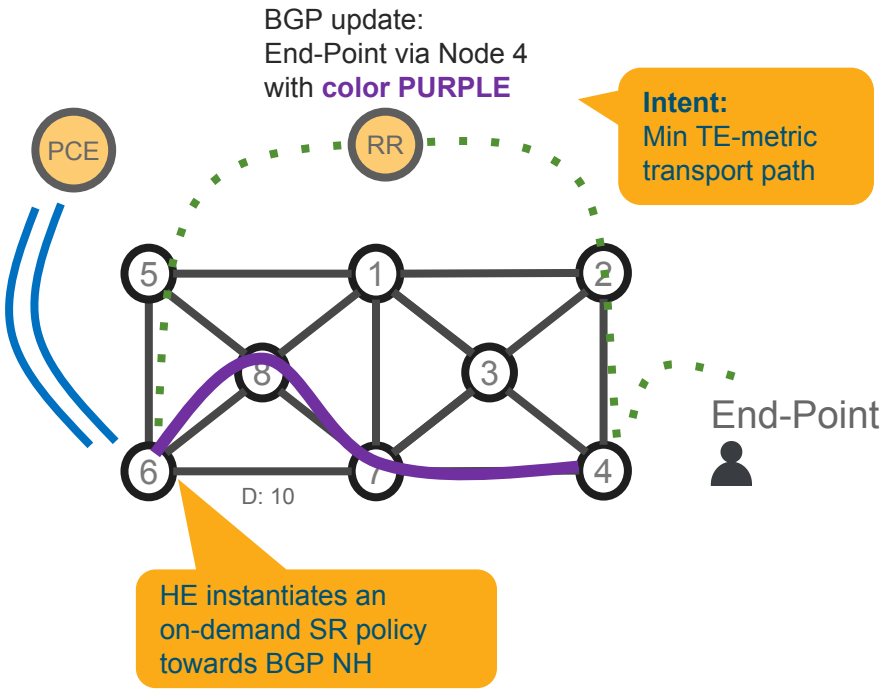
Benefits

Intent-based

SLA-aware BGP service
Decoupled service and transport provisioning

Scalable

No a-priori full-mesh of connectivity



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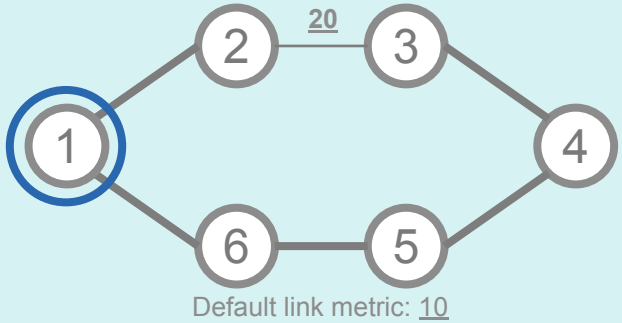
SR Policy – Configuration Example

On Node1:

```
segment-routing
 traffic-eng
  policy POLICY1
   color 20 end-point ipv4 1.1.1.4
   binding-sid mpls 1000
   candidate-paths
    preference 100
     dynamic
      metric type te
     constraints
      affinity
       exclude-any color red
    !
    preference 200
     explicit segment-list SIDLIST1
  !
  segment-list name SIDLIST1
   index 10 mpls label 16002
   index 20 mpls label 30203
   index 30 mpls label 16004
```

SRTE

SR Policy



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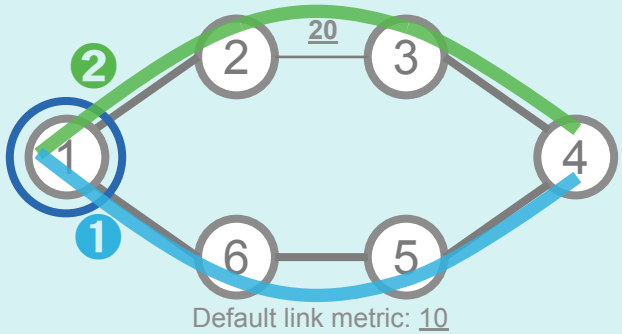
User-defined name

Color and End-point

Binding-SID

Local Candidate Paths



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Path preference 100

Dynamic path

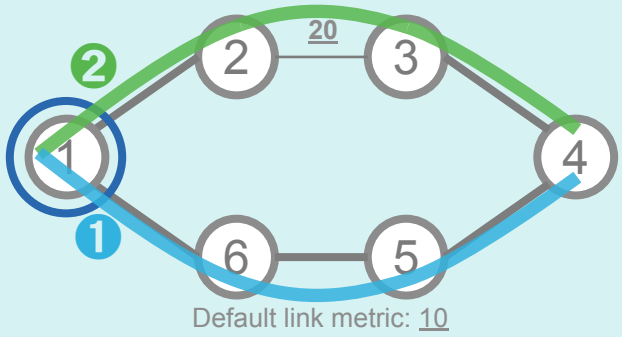
Opt. Obj.: TE metric

Constraint

Path preference 200

Explicit SID-list1

SID-list1



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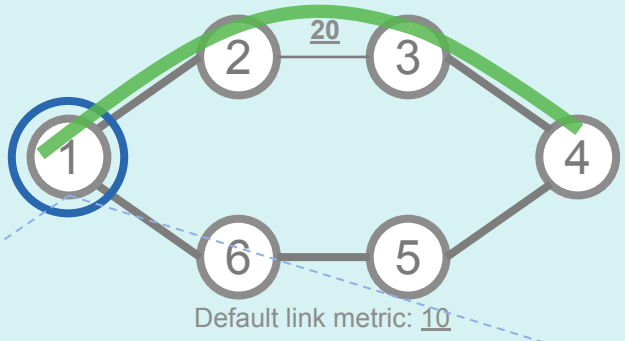
```

Selected Path:
- Valid Path
- Highest Pref value



FIB @ head-end Node1
Incoming label: **1000**
Action: pop and push <**16002**, **30203**, **16004**>

Intent-based Automated Steering

What it Does

Automatic Traffic Steering

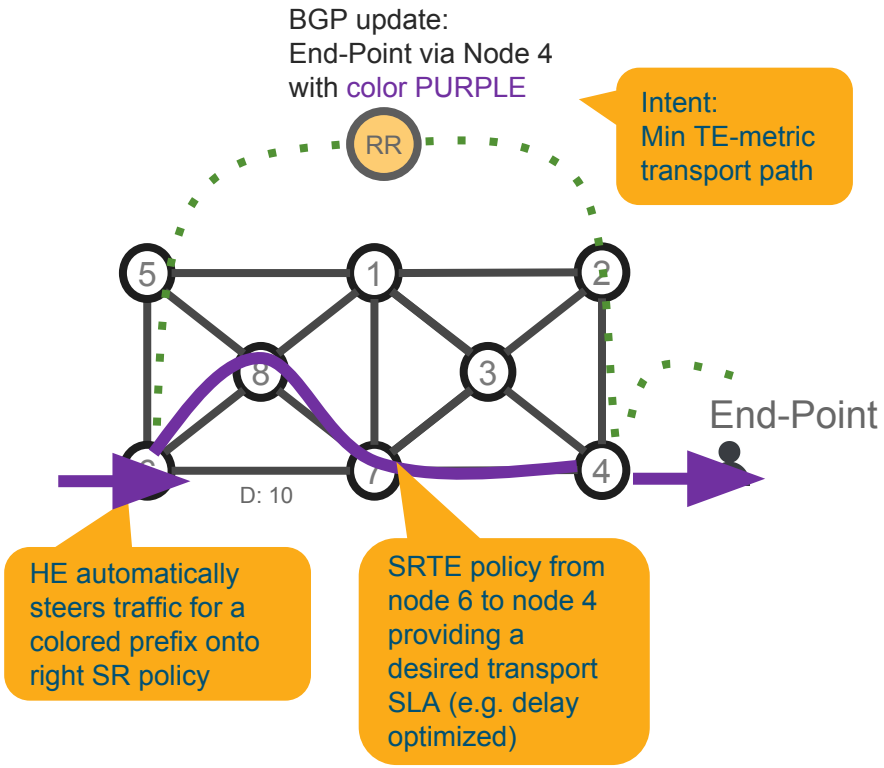
By default, services are automatically steered on the right SR policy based on next-hop and color of the service route. This can be disabled in the SR Policy.

Delivers across the unified SR Fabric the SLA requested by the service

Benefits

Simplicity, automation and scale

No traditional policy-routing
Uses existing ODN SR Policies
No performance penalty



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Flex-Algo: SR IGP Flexible Algorithms

What it Does

Customized IGP algorithms defined by operator for intent-based instantiation of traffic Engineering

Minimization of metrics: IGP, delay.
Exclusion of properties: link-affinity, SRLG.

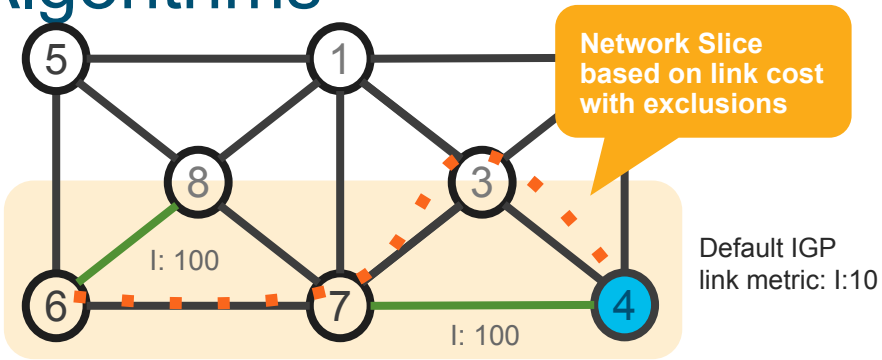
Benefits

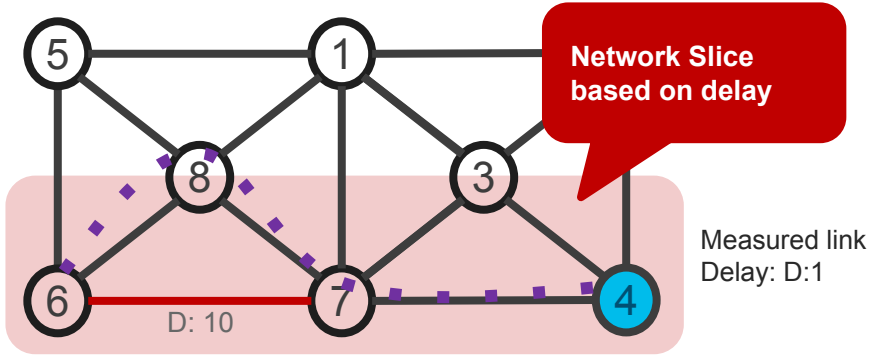
Simplicity and Automation

IGP-computed TE-path from anywhere to anywhere
Sub-50msec protection (TILFA) optimized per Flex-Algorithm plane.

Scalability

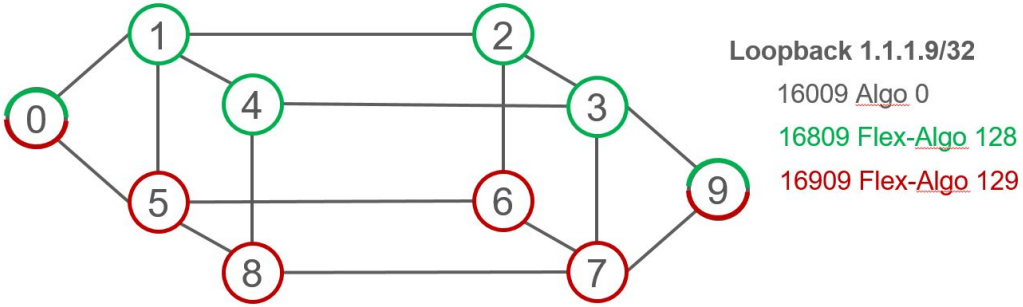
Single SID (instead of label stack) to enforce TE path.
Single prefix segment can participate in many Flex-Algos.



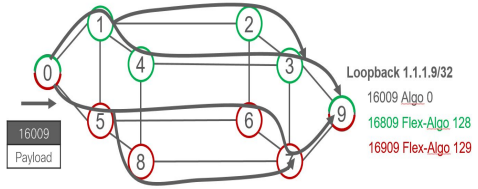


Same destination, different colour, different optimisation goals and constraints

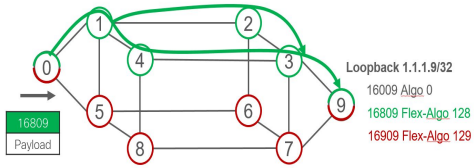
Flex-Algo: Multi-Plane Topologies



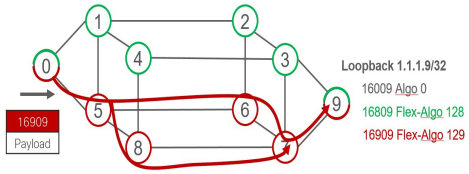
- All the nodes support Algo 0: minimize IGP metric
- Green nodes also support 128: minimize IGP metric
- Red nodes also support 129: minimize Delay



- Path to Node 9 across Algo 0

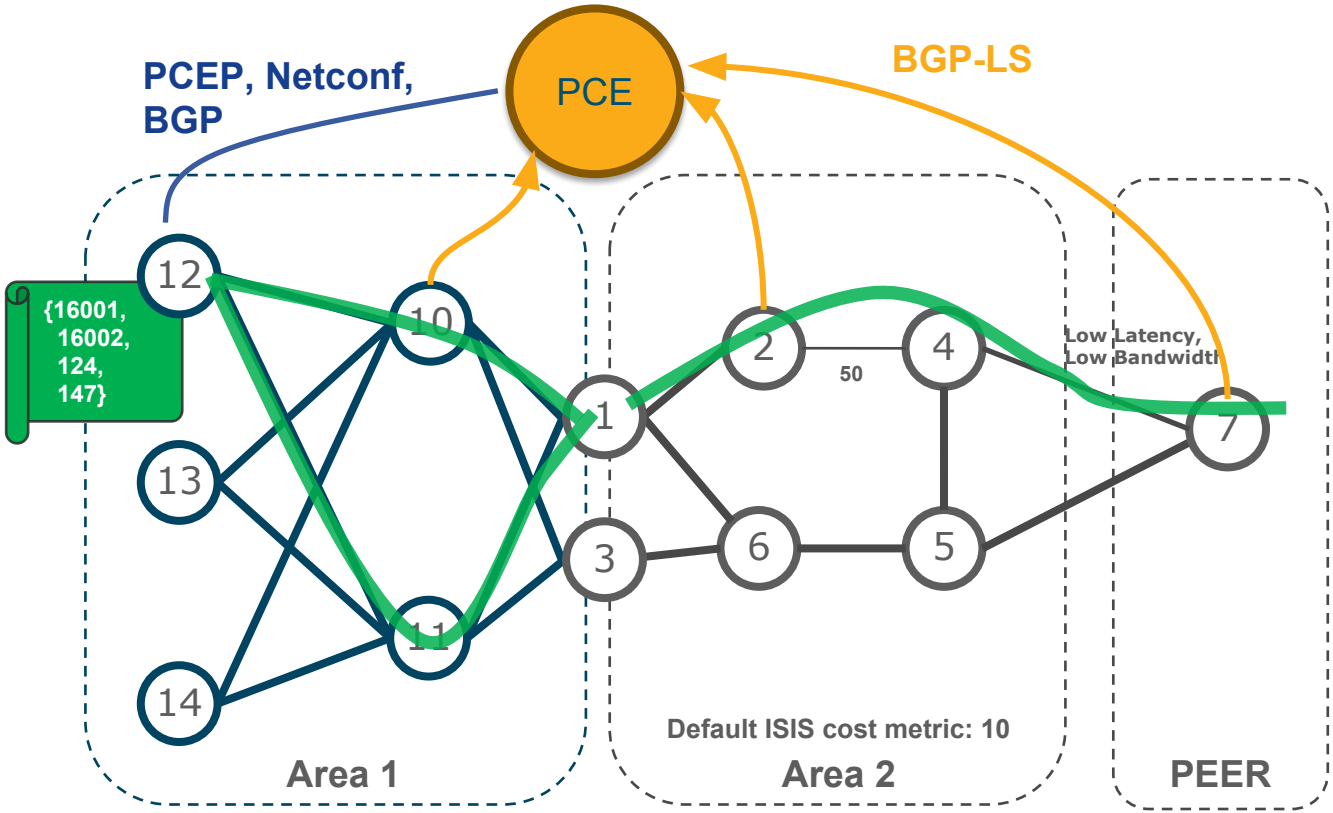


- Path to Node 9 across Flex-Algo 128



- Path to Node 9 across Flex-Algo 129

Automation for Segment Routing Traffic Engineering



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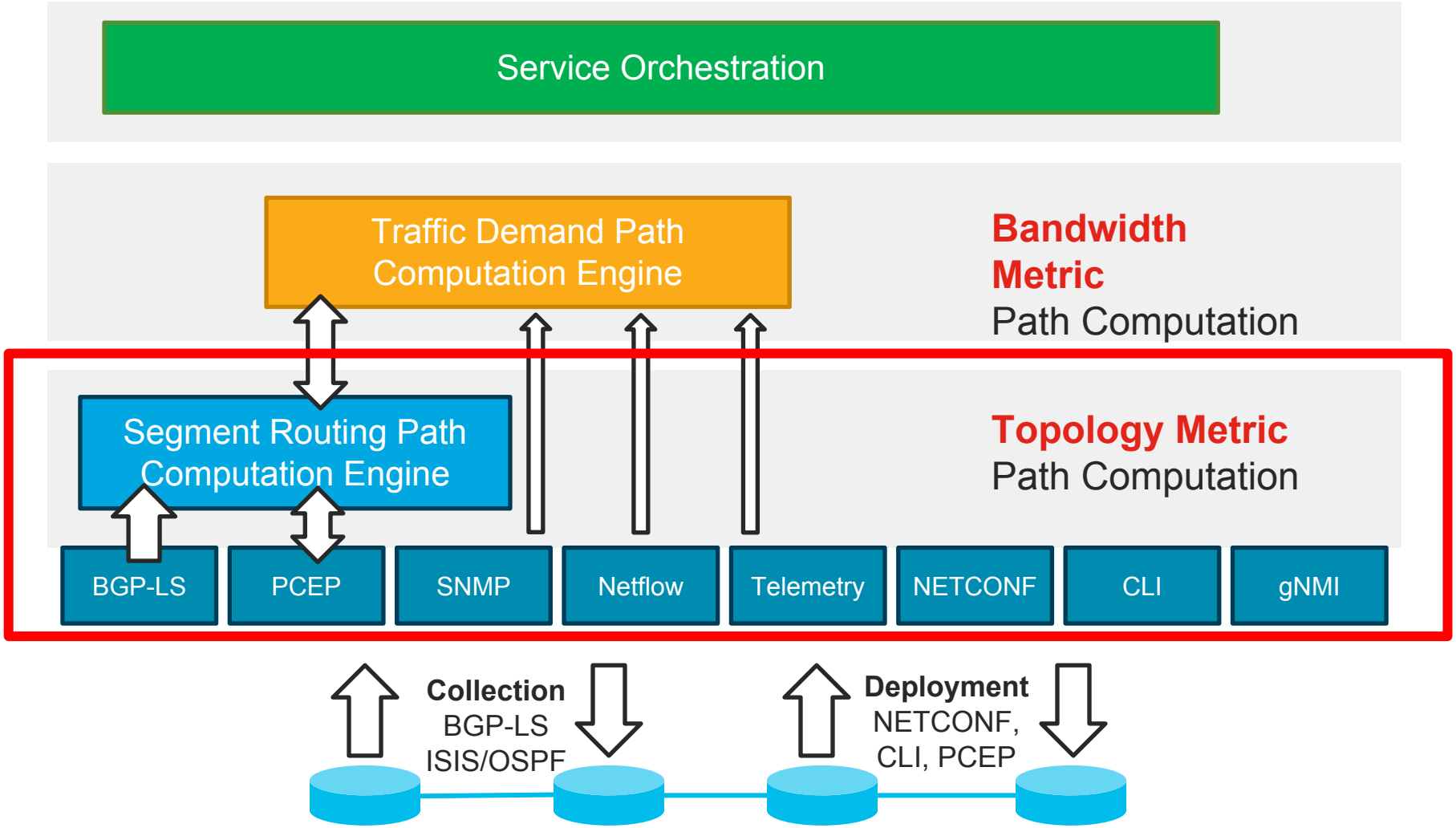
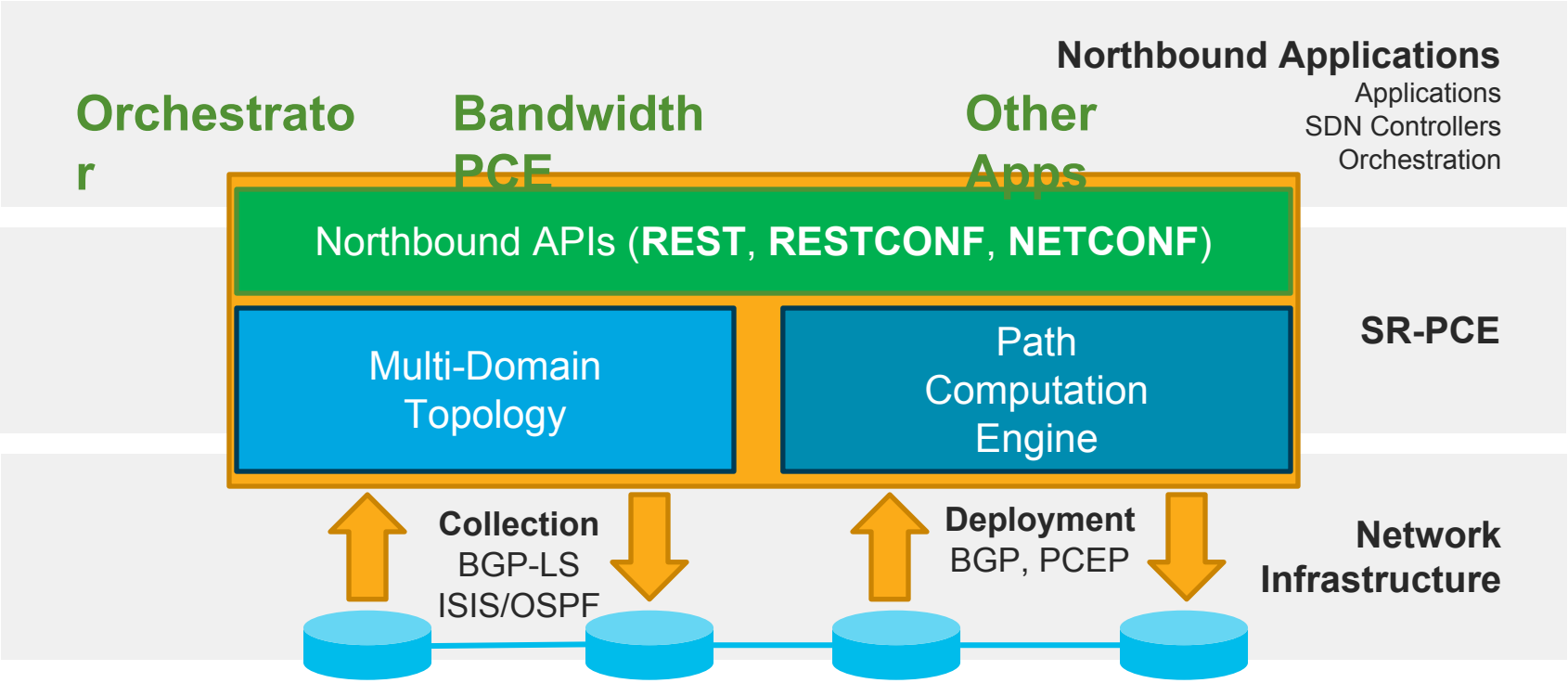
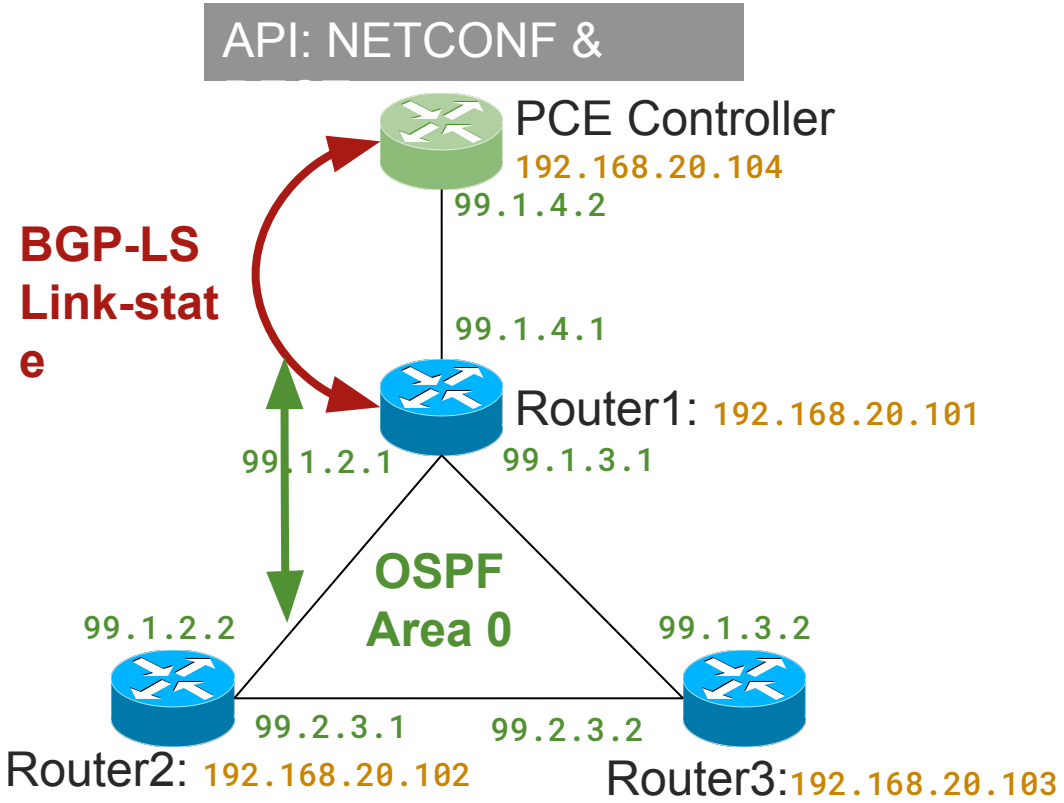

Service Orchestration
Traffic Demand Path Computation Engine
Bandwidth Metric
Path Computation
Segment Routing Path Computation Engine
Topology Metric
Path Computation
BGP-LS
PCEP
SNMP
Netflow
Telemetry
NETCONF
CLI
gNMI
Collection
BGP-LS
ISIS/OSPF
Deployment
NETCONF,
CLI, PCEP

Segment Routing Path Computation Engine



Demo Environment



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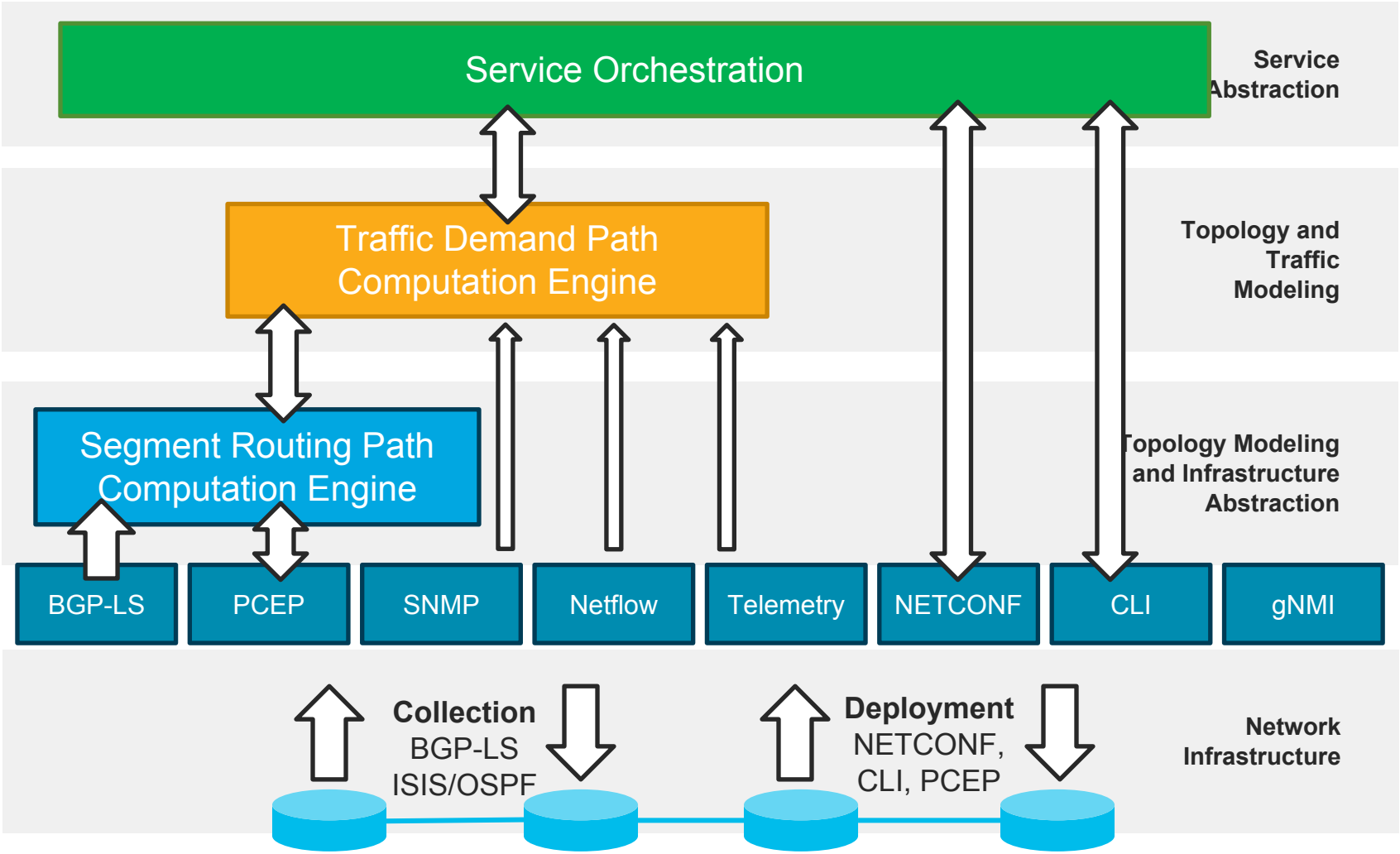
Service Orchestration
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